Causal Inference

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1 Introduction / Abstract

In Causal Inference, the selection of covariates play a crucial role in determining the causal effect. In this simulation study on observational data, we review the effect of variable selection using different types of covariates, to effectively answer the question `"Which variables exactly should you control for?"`

2 Data generating process

2.1 Covariate generation

The data was generated according to Setup 1 for the continuous case, in accordance to the article. The DGP used the following parameter values:

Table 1: Parameter values

Category	Parameter	Value
Independent covariates	$X_1, X_2, X_4, X_6, X_7, U$	N(0, 1)
Dependent variable	X_3	$U + \epsilon_3$
Dependent variable	X_5	$X_4 + \epsilon_5$
Dependent variable	X_8	$X_2 + X_6 + \epsilon_8$
Error terms	$\epsilon_3, \epsilon_5, \epsilon_8, \epsilon_Y$	N(0, 1)
True causal effect	τ_{true}	0.5

Table 1 outlines the characteristics of the data generation process of the covariate generation. As seen in Table 1, the independent covariates $(X_1, X_2, X_4, X_6, X_7)$ and an unobserved (latent) variable U, were generated from a standard normal distribution $N(0, 1)$. Then, three dependent covariates $(X3, X5, X8)$ were generated as linear functions of the independent covariates (see Table 1) to simulate causal relationships, and then standardized. Here, X_3 is used as a 'proxy' and models the unobserved (latent) variable U, X_5 is a function of X_4 and X_8 works as a collider by X_2 and X_6. The true causal effect is finally estimated to 0.5 to evaluate the efficiency of the different estimation methods. The DGP was implemented in R for 1000 simulated datasets, each with a sample size of 500.

2.2 Treatment and outcome models

Next, the assignment mechanism through the treatment and outcome models are defined:

A binary treatment variable W was generated using the following logistic function to calculate the propensity score:

$$P(W = 1) = \frac{1}{1 + \exp(-(X_1 + X_2 + X_3 + X_4))}$$

The X variables all serve different purposes for the experiment. X_1 is a pure treatment variable that only affects who gets the treatment but has zero influence on the outcome, which could also be seen in figure 1 below. X_2 influences treatment, but also X_8, where X_8 will act as a collider through both X_2 and X_6. X_3 affects the treatment, but X_3 itself is affected by an unobserved latent variable, thus creating a backdoor path which can only be blocked by adjusting for X_3. Backdoor paths are indirect and non-casual routes, as seen in the DAG. We want to avoid these since these paths create confounding. X_4 acts as a confounder and affects the treatment, but also has an influence on X_5, which affects the outcome.

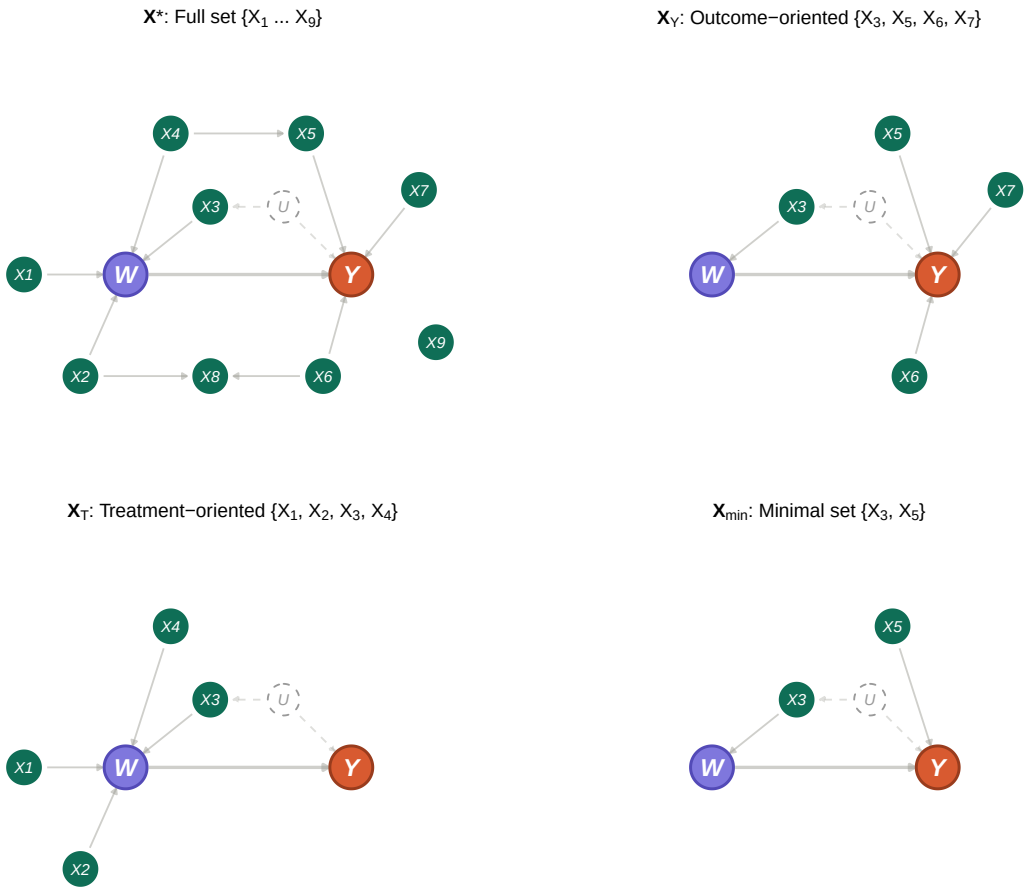


Figure 1: DAG for the data-generating process

The outcome was then generated as:

$$Y = U + X_5 + X_6 + X_7 + W \cdot \tau_{true} + \epsilon_Y$$

These X variables were also chosen for a specific reason. X_7 only has an effect on the outcome, but this time does not influence the treatment. X_5 affects the outcome but is also affected by X_4, which influences treatment, thus creating another backdoor path. X_6 influences the outcome but acts as a collider for X_8 as previously mentioned. Finally, U directly influences the outcome but are also affected by X_3, which influences treatment, thus creating a backdoor path.

2.3 Target sets

Studying the DAG gives us the 4 target sets that will be used to evaluate the 4 different methods to estimate the ACE (Witte and Didelez 2019).

Table 2: Target sets

Selection set	**Variables**
Full target set (X*)	$X_1, X_2, X_3, X_4, X_5, X_6, X_7, X_8, X_9$
Minimal target set (Xmin)	X_3, X_5
Outcome oriented target set (XY)	X_3, X_5, X_6, X_7
Treatment oriented target set(XT)	X_1, X_2, X_3, X_4

- **Full Target Set:** This set includes all measured covariates in the dataset, no matter which connections we have.
- **Minimal Target Set:** The minimal set is the smallest set that blocks all backdoor paths. Looking at the DAG in Figure 1, there are, as we discussed previously, three backdoor paths;

  1. $W \leftarrow X_4 \rightarrow X_5 \rightarrow Y$
  2. $W \leftarrow X_3 \leftarrow U \rightarrow Y$

3. $W \leftarrow X_2 \rightarrow X_8 \leftarrow X_6 \rightarrow Y$

Since X_8 is a collider, this path will already be blocked. Since X_3 is the only variable able to block the backdoor path from the unobserved variable U, this one has to be chosen. Then we can choose between X_4 and X_5 to block the remaining backdoor path. There are therefore two possible minimal sets to block the backdoor paths, these are (X_3, X_4) or (X_3, X_5). This study chooses the second one for the minimal target set for efficiency since X_5 contains all the information in X_4 ($X_5 = X_4 + \epsilon_5$).

- **Outcome-oriented Target Set:** The outcome-oriented target set focuses on all direct predictors of the outcome, which in the DAG is X_3 (as a proxy for U), X_5, X_6, X_7.
- **Treatment-oriented Target Set:** The treatment-oriented target set instead focuses on all direct predictors of treatment, which in this dataset is X_1, X_2, X_3, X_4.

3 Results

The different targets will estimate the ACE using four different estimation methods: linear regression, matching, propensity score matching and doubly robust estimation.

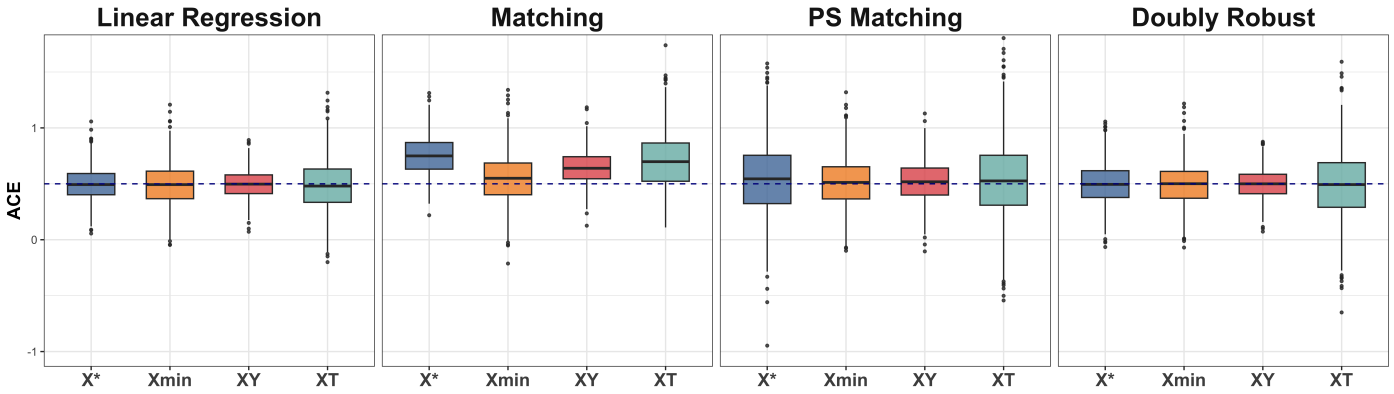


Figure 2: ACE adjusted for target sets across linear regression, matching, propensity score (PS) matching, and doubly robust estimation. The true ACE is 0.5 (dashed line)

Table 3: Estimation Performance Metrics by Method and Variable Set

Method	Variable Set	Bias	Empirical SE	RMSE
Linear Regression	X*	-0.0032	0.1441	0.1440
	Xmin	-0.0053	0.1827	0.1827
	XT	-0.0131	0.2261	0.2264
	XY	-0.0012	0.1252	0.1252
Matching	X*	0.2496	0.1672	0.3004
	Xmin	0.0446	0.2112	0.2157
	XT	0.1983	0.2482	0.3176
	XY	0.1423	0.1428	0.2016
PS Matching	X*	0.0405	0.3241	0.3264
	Xmin	0.0131	0.2243	0.2246
	XT	0.0300	0.3505	0.3516
	XY	0.0171	0.1812	0.1819
Doubly Robust	X*	-0.0013	0.1844	0.1843
	Xmin	-0.0059	0.1865	0.1865
	XT	-0.0150	0.2986	0.2988
	XY	-0.0010	0.1285	0.1284

First, the study estimates the average causal effect (ACE) using linear regression. If the model is correctly specified, the treatment coefficient then represents the ACE. Looking at figure 2, it can be seen that using linear regression resulted in unbiased estimators for all target sets.

The second method chosen is covariate matching, which works by matching treated and control observations with similar covariates (Huber 2023). The matching takes the Mahalanobis distance between the covariates and then calculates the ACE estimates from these similar observations. As seen in figure 2 and table 3 the quality of matching gradually decreases as more variables need to be matched on, which leads to biased estimators for all sets except the minimal one, with the largest bias observed for the full target set (Witte and Didelez 2019).

Propensity score (PS) Matching is a special form of matching where units are instead matched based on their propensity scores (Waernbaum 2026b). Here, it can be noted how collapsing all covariates into a single dimension resulted in unbiased ACE estimates for all target sets, solving the dimensionality issue observed in the regular matching. The variance is the smallest for the outcome target set and the highest for the full target-oriented and the treatment-oriented set, as adjusting for strong predictors tends to improve precision and reduce standard errors, while including too many variables only affecting treatment tends to lead to a higher variance and lower precision (Huber 2023).

The final estimation method is the doubly robust estimator, which combines an outcome regression model with a propensity score weighting model. By combining these models, the doubly robust estimator is able to give unbiased estimators for all target sets while also providing quite low variances. However, the treatment-oriented target set again has one of the highest variances, caused by including too many treatment-predicted variables that are not needed for confounded (Witte and Didelez 2019).

4 Discussion of Results

4.1 Compare the SE of the minimal set vs. the outcome-oriented set. Why does outcome predictors reduce the variance of the treatment effect estimate even though they are not true confounders?

Table 4: Empirical SE: Minimal Set ($X_{\min}$) vs. Outcome-Oriented Set (X_Y)

Method	**Empirical SE ($X_{\min}$)**	**Empirical SE (X_Y)**
Linear Regression	0.1827	0.1252
Matching	0.2112	0.1428
PS Matching	0.2243	0.1812
Doubly Robust	0.1865	0.1285

As shown in the table above, the outcome-oriented set (X_Y) consistently yields lower standard errors than the minimal set (X_{min}) across all four estimation methods, with improvements ranging from approximately 19% to 32%.

Even if a covariate does not confound the treatment–outcome relationship, including it in the adjustment set can still improve estimation efficiency if it is predictive of the outcome (Imbens and Rubin 2015). The goal of any causal estimator is to isolate the 'true' average treatment effect of W on Y. Covariates that are strongly associated with the outcome would help isolate the effect of W more precisely, thus removing the residual 'noise'.

A core foundation of Causal theory states that confounders should be accounted for in order to yield optimal results. The minimal sufficient set X_{min} satisfies this condition minimally, as well as the superset containing the outcome-oriented set X_Y. In real-world applications, these outcome-oriented variables can cause unexplained swings in the outcome that are not directly related to the treatment. Removing these factors causes the noise to disappear, allowing the treatment signal to become far more confident and hence reporting a tighter margin of error (lower SE).

To understand this from a statistical perspective, we can start by looking at the semiparametric efficiency bound of the variance (Waernbaum 2026a):

$$V_{sp}^{eff} = E_{sp}\left[\frac{\sigma_c^2(x)}{1-e(X_i)} + \frac{\sigma_t^2(x)}{e(X_i)} + (\tau_{sp}(X_i) - \tau_{sp})^2\right]$$

The third term $(\tau_{sp}(X_i) - \tau_{sp})^2$ represents the variance of the treatment effect. However, since the treatment effect is constant, this term vanish to zero. This results in the efficiency bound simplying to:

$$V_{cond}^{eff} = E_{sp}\left[\frac{\sigma_c^2(x)}{1-e(X_i)} + \frac{\sigma_t^2(x)}{e(X_i)}\right]$$

The bound therefore, depends on two types of quantities:

- The conditional outcome variances: $\sigma_c^2(x)$ and $\sigma_t^2(x)$, which appear in the numerator.
- The propensity score: $e(X_i)$, which appears in the denominator.

Since these additional outcome-oriented covariates (X_6, X_7) are independent of the treatment W, $e(X_{min}, X_6, X_7) = e(X_{min})$. Hence, we focus on the conditional outcome variances and compare them across both sets: $\sigma^2(X_{min})$ and $\sigma^2(X_Y)$.
To see exactly why $\sigma^2(X_Y) < \sigma^2(X_{min})$, we compare the error terms of their regression models.

First, we define the min model using X_{min} set. The error term here is ϵ_{min}:

$$\boldsymbol{Y = \tau W + \beta X_{min} + \epsilon_{min}}$$

Next, we run the outcome-oriented model using the X_Y set, which includes the remaining outcome-oriented variables $X^*_{out} = (X_6, X_7)$. The error term here is ϵ_{out}:

$$\boldsymbol{Y = \tau W + \alpha X_{min} + \beta^* X^*_{out} + \epsilon_{out}}$$

Because X_6 and X_7 directly influence the outcome, they help to explain and hence reduce the previously unexplained portion of the variance. Additionally, since X^*_{out} is uncorrelated with X_{min} and W, the coefficients for W and X_{min} remain consistent between the models (per the FWL theorem), ie. $\alpha = \beta$.

Hence, the relationship between the two error terms simplifies to:

$$\boldsymbol{\epsilon_{min} = \beta^* X^*_{out} + \epsilon_{out}}$$

Taking the Variance;

$$\begin{aligned}
\text{Var}(\boldsymbol{\epsilon}_{\text{min}}) &= \text{Var}(\mathbf{X}^*_{\text{out}}\boldsymbol{\beta}^* + \boldsymbol{\epsilon}_{\text{out}}) \\
&= \text{Var}(\mathbf{X}^*_{\text{out}}\boldsymbol{\beta}^*) + \text{Var}(\boldsymbol{\epsilon}_{\text{out}}) + 2\text{Cov}(\mathbf{X}^*_{\text{out}}\boldsymbol{\beta}^*, \boldsymbol{\epsilon}_{\text{out}}) \\
&= \text{Var}(\mathbf{X}^*_{\text{out}}\boldsymbol{\beta}^*) + \text{Var}(\boldsymbol{\epsilon}_{\text{out}}) \quad (\text{since } \text{Cov}(\mathbf{X}^*_{\text{out}}\boldsymbol{\beta}^*, \boldsymbol{\epsilon}_{\text{out}}) = 0)
\end{aligned}$$

Finally, since variance is strictly greater than 0, we can conclude that $\text{Var}(\boldsymbol{\epsilon}_{\text{out}}) < \text{Var}(\boldsymbol{\epsilon}_{\text{min}}) \implies \text{Var}(\boldsymbol{\epsilon}_{\text{Y}}) < \text{Var}(\boldsymbol{\epsilon}_{\text{min}}) \implies \sigma^2(X_Y) < \sigma^2(X_{min})$.

4.2 Discuss the impact of X_1 (the instrument). Does adding X_1 to the model improve the precision of the ATE estimate in this OLS setting, or does it stay the same?

As described in the DGP, X_1 is a treatment-only covariate with no influence on the outcome. The inclusion of terms with little-to-no predictive power for the outcome tends to result in an inflation of variance. This causes the OLS model to overfit, and hence create patterns out of the error terms rather than identifying true structural relationships. This causes the estimates in the OLS regression to become imprecise, thus worsening the precision of the ACE estimate. (Huber 2023).

Table 5: Estimation Performance Metrics: Adding X_1 to Minimal and Outcome Sets

OLS	**Empirical SE**	**RMSE**
Minimal Set		
X_{min}	0.1827	0.1827
$X_{\text{min}} + X_1$	0.1927	0.1928
Difference	+0.0100	+0.0101
Outcome Set		
X_Y	0.1252	0.1252
$X_Y + X_1$	0.1313	0.1312
Difference	+0.0061	+0.0060

This problem can be illustrated in Table 5. The outcome-oriented set was the most precise in OLS, while the minimal set is the minimum sufficient for unbiasedness. Including X_1 in these sets shows how introducing a variable with no predictive power negatively affects precision. Adding X_1 to the X_{min} set resulted in a larger empirical standard error and RMSE as the OLS regression captures patterns in the error terms, introducing unnecessary noise without reducing the residual sum of squares (Huber 2023). The same could be seen when adding X_1 to the outcome-oriented set. Both the empirical standard error and RMSE increased. This confirms the theory that introducing too many terms that have no or very small influence on the outcome (Y) can often result in a less precise estimate.

5 References

References

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